The National Higher School of Artificial Intelligence
Introduction to Artificial Intelligence
Midterm Exam 18/05/2023
Duration: 1 hour 45 minutes

Exercise 1: 3 points (0.5, 0.5, 0.5, 0.75, 0.75)

Answer and give a brief explanation to the following questions.

1. A definition of Artificial Intelligence is that it is "the study of how to make computers do things at which, at the moment, people are better". Which school of AI does this definition point to? Explain.

*This is the **Acting Humanly** school of AI which considers that Intelligent systems are supposed to mimic humans' way of **thinking** but also be able to **do** things which involve action (in addition to thinking).*

2. Another definition of Artificial Intelligence is that it is "a field of study that tries to explain and emulate intelligent behaviour in terms of computational processes". Which school of AI does this definition point to? Explain.

*This is the **Thinking Humanly** school of AI which considers that Intelligent systems are supposed to mimic humans' way of **thinking** (only) i.e. in terms of "computational" processes.*

3. What is cognitive science?

Cognitive Science is the science which brings together computer science models from AI and experimental techniques from psychology to construct precise and testable theories of the human mind.

4. Suppose that we want to represent a node of a search tree as a record. What information would be needed in such a node to implement the general search algorithm?

Such a node should contain at least the following information:
- *STATE: the state in the state space to which the node corresponds;*
- *PARENT node: the node in the search tree that generated this node;*
- *ACTION: the action that was applied to the parent to generate the node;*
- *PATH-COST: the cost, traditionally denoted by g(n), of the path from the initial state to the node, as indicated by the parent pointers.*

5. Suppose we are using a path cost function with some search technique. Is it possible that the cost of a path decreases at some point going from the root downward? If so, say when. Could you give an example of a problem where this would happen?

Yes; depending on the problem, the cost can decrease from the root downward.
Example: Suppose a problem where the agent is penalised for an action considered bad. For instance, cost_function(to state n) = path_cost(from root to state n) + cost(last action that brought to state n) where

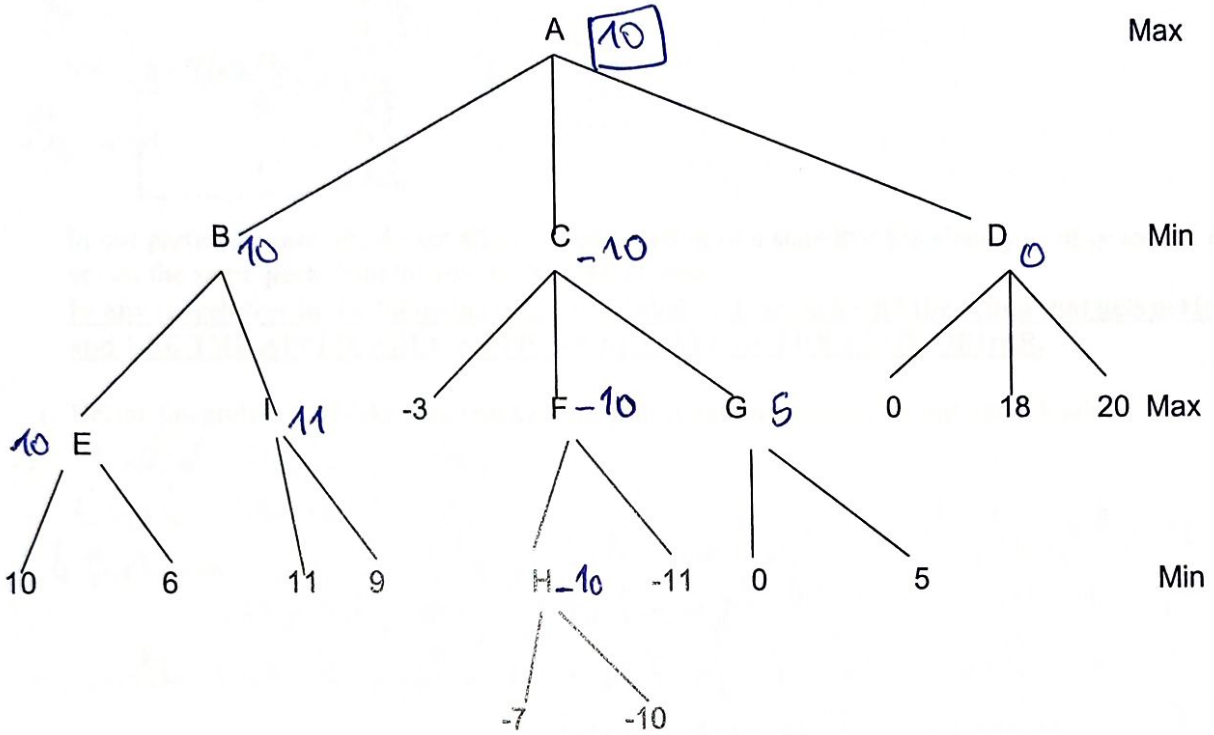
$$Cost(any\ action) = \begin{cases} +10 & good\ move \\ +3 & move\ ok \\ -5 & bad\ move \end{cases}$$

Obviously, we assume that there are criteria for evaluating the quality of the action/move

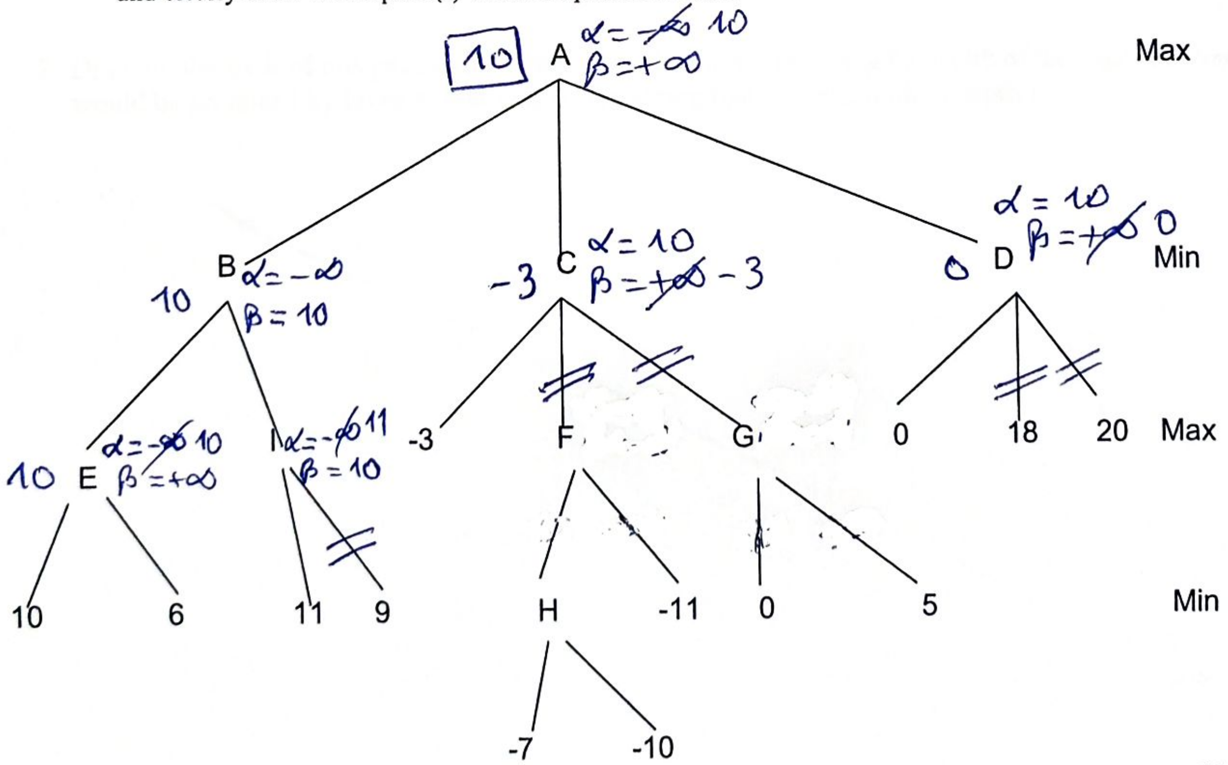
Exercise 2: **4 points**

Consider the following 4-ply game.

1. Pass the values from the leaf nodes back up reflecting the work of the minimax procedure.



2. Use the minimax procedure with the alpha-beta cutoffs, showing ALL the values as per the algorithm and clearly show which parts(s) would be pruned if at all.



Exercise 3: **7 points** (1, 1, 3, 1, 1)

Consider the 8-puzzle problem where we want to move from the configuration

Let us denote the states

$$S_0 = \begin{bmatrix} 2 & 8 & 3 \\ 1 & 6 & 4 \\ 7 & & 5 \end{bmatrix}$$

to the configuration

$$S_G = \begin{bmatrix} 1 & 2 & 3 \\ 8 & & 4 \\ 7 & 6 & 5 \end{bmatrix}$$

In our particular case, we do not allow the **generation** of a state that has already been generated higher up, on the **same path** from the root to the current state.
<u>In any search tree in the following questions, label each branch with the action that gets performed and USE THE APPLICABLE ACTIONS ALWAYS IN THE SAME ORDER.</u>

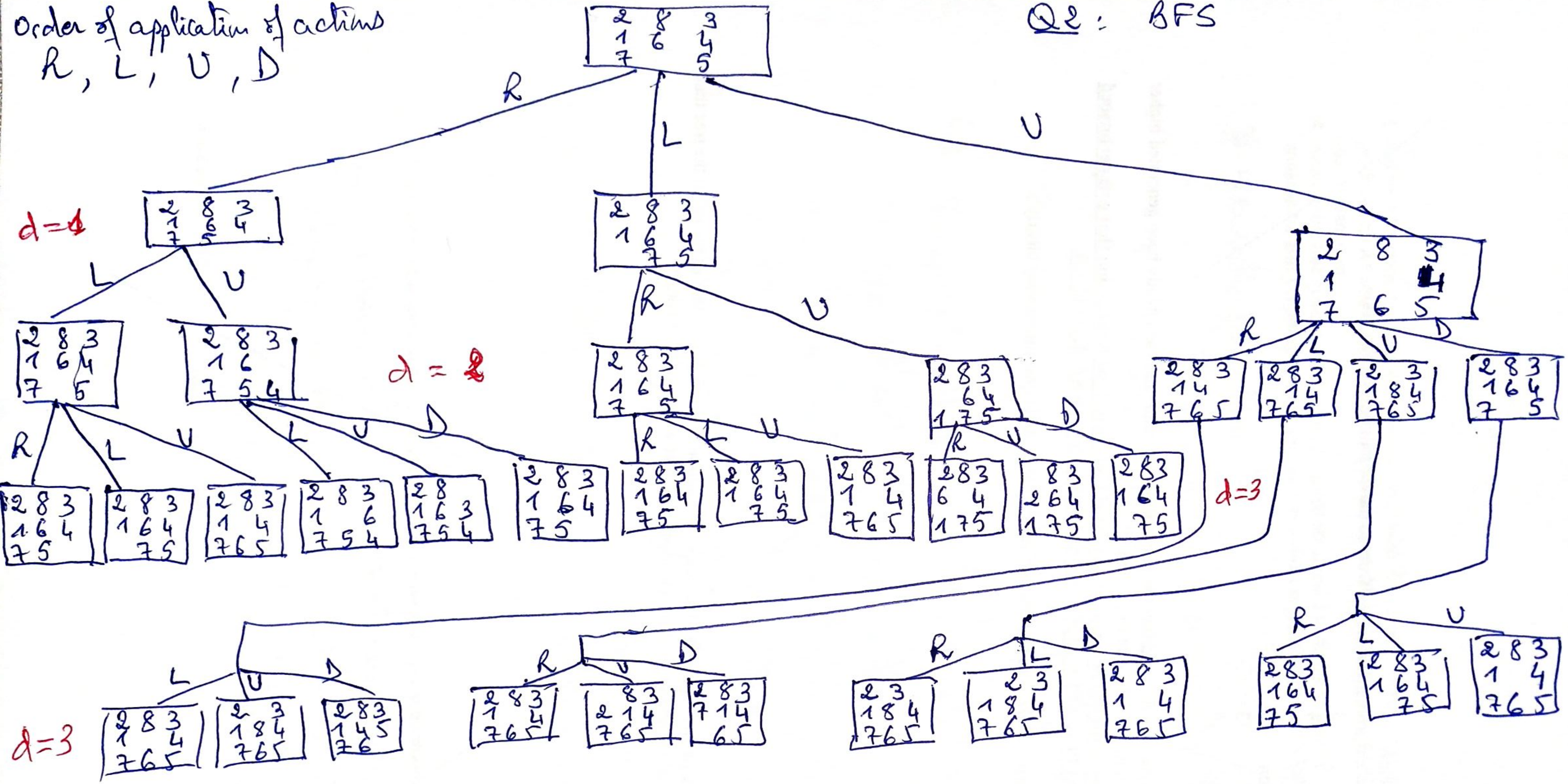
1. Define the problem. (Make sure you explain your notation and conventions very clearly.)

- Initial state: S_0
- Goal state: S_G
- 4 Actions: move blank tile right/left/up/down allowed; they will be applied in this order if applicable.
- path cost function: cost of 1 for each move. Sum up all the individual moves.

2. Draw on the back of this page in the landscape orientation (i.e. using the width of the page) the tree that would be produced by breadth-first search, supposing that we set a limit to depth 3.

Order of application of actions
R, L, U, D

Q2: BFS



3. Define a heuristic function that could be used with A* search. Draw on the back of this page the tree produced by A* search using your heuristic function. Make sure you indicate above each state in the tree the relevant cost.
4. Analyse the tree generated as a result of questions 2 and 3 to define an evaluation function that would be suitable for Hill Climbing. State it and use it to solve the problem (with the Hill Climbing strategy).

3) Heuristic function : h

The number of moves to take each tile to its destination,

The cost function $f(n)$ in A* is $g(n) + h(n)$, where g is the number of moves from the root to this state m.

4) Comparing the trees generated for questions 2 and 3, we see that the heuristic function used in Q3 has helped go "straight to the goal" in A*, avoiding all the going back to previously visited states. So if the same heuristic is used as an evaluation function for Hill Climbing, we will be able to avoid any plateaux,...

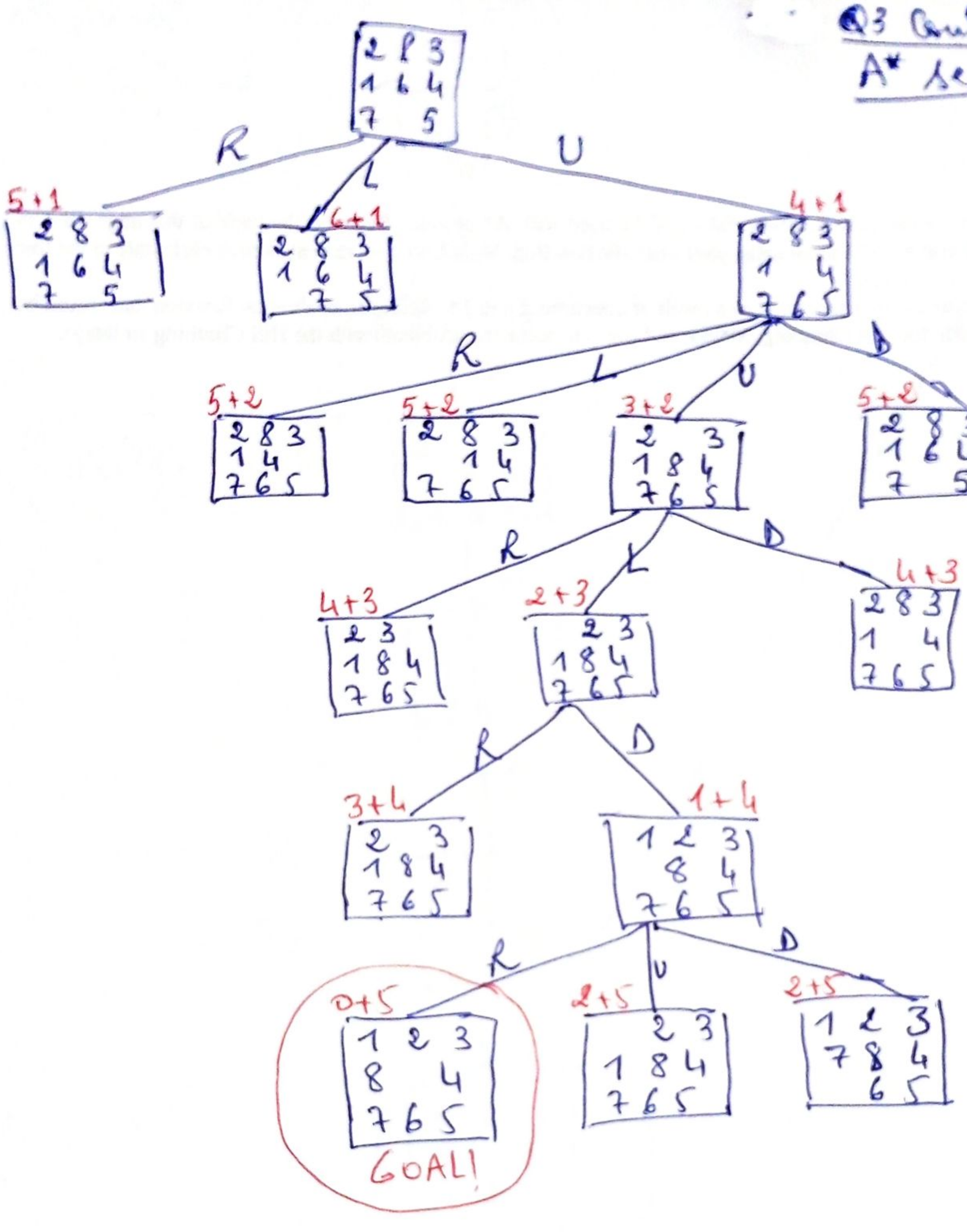
5. Briefly say what you can conclude comparing the results you got at questions 2, 3, and 4.

→ The heuristic function that was used has avoided all the loops that appear due to BFS, So A* and HC are much better in this case than BFS.

→ The local search strategy HC is more effective than A* in this case.

Q3 Continued

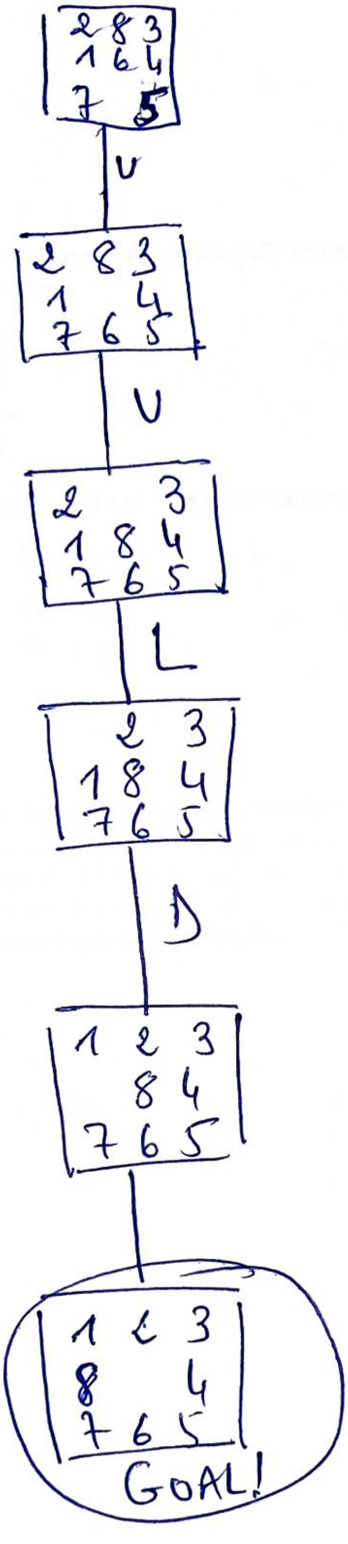
A* search



Total Cost : 5

Hill climbing

Question 4 Continued



Exercise 4: **6 points** (1,5)

Consider the following cryptarithmetic problem where all the letters take values between 0 and 9 and each letter must be different from all others.

MEN
ARE

WISE

1. State the CSP problem precisely and define the domains entirely.

$N+E = E + 10C_1$
$C_1+E+R = S + 10C_2$
$C_2+M+A = I + 10C_3$
$C_3 = W$

$W, C_1, C_2, C_3 \in \{0,1\}$
all other variables $\in [0..9]$

2. Using the studied heuristics appropriately and explaining which one is selected at any point, or which arithmetic/propagation is done, show on a search tree (making clear any backtracking that occurs) whether the problem has a solution or not. If it does have one, give it. Each node in the search tree will show all the applicable constraints at that point. (In order not to get lost/confused, make sure you apply the heuristics systematically as explained in the lectures.) Do not use any intelligent backtracking.

See next page...

$N + E = E + 10 C_1$
$C_1 + E + R = S + 10 C_2$
$C_2 + M + A = I + 10 C_3$
$C_3 = W$

$C_1, C_2, C_3, W \in \{0,1\}$
$N, E, R, M, A, I \in [0..9]$

Arithmetics + Propagation

$N = 10 C_1 \Rightarrow N = 0 = C_1$
$C_1 + E + R = S + 10 C_2$
$C_2 + M + A = I + 10 C_3$
$C_3 = W$

$C_2, C_3, W \in \{0,1\}$
$E, S, R, M, A, I \in [1..9]$

MRV

$W = C_3 = 1$

$W = C_3 = 0$

$C_1 + E + R = S + 10 C_2$
$C_2 + M + A = I + 10$

$N = C_1 = 0$
$W = C_3 = 1$
$C_2 \in \{0,1\}$
all others $\in [2..9]$

MRV $\Rightarrow C_2$
Try $C_2 = 0$

$C_2 = 1$

$E + R = S$
$M + A = I + 10$

$N = C_1 = C_2 = 0$
$W = C_3 = 1$
$E, R, M, A, I, S \in [2..9]$

MRV all same; DH all same
select E; Try $E = 2$

$2 + R = S$
$M + A = I + 10$

$R \in [3..9]$; $S \in [5..9]$
$M, A, I \in [3..9]$

MRV select S
Try $S = 5$

$R = 3$
$M + A = I + 10$

$M, A, I \in \{4, 6, 7, 8, 9\}$

MRV and DH same
Select A; Try $A = 4$

$M = I + 6$
$I \in \{4, 6, 7, 8, 9\}$
$\Rightarrow M \geq 10$

Impossible

Try $A = 6$

$M = I + 4$

$I \in \{4, 7, 8, 9\}$
$M \in \{7, 8, 9\}$

MRV $\Rightarrow M$
Try $M = 7$

$M = 7 \Rightarrow I = 3$

Impossible

Try $M = 8$

$M = 8 \Rightarrow I = 4$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  M E N       8 2 0
+ A R E     + 6 3 2
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W I S E     1 4 5 2
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